Agentic AI Autonomous Cold-Chain Architecture

Next-generation sustainability via 1.2s latency thermal forecasting and autonomous decision agents.

Autonomous Workflow: Sensing to Action

SENSING

Input Layer: Real-time IoT data ingestion (Temperature, Humidity, GPS) from edge sensors.

LSTM ENGINE

Intelligence: Time-series forecasting with 1.2s latency. Predicts breaches 4 hours in advance.

AGENTIC LOGIC

Decision Core: Risk evaluation via kinetic degradation modeling.

$$k = Ae^{-\frac{E_a}{RT}}$$

ACTION

Execution: Trigger autonomous reroute to nearest MFC. Result: 66% Spoilage Reduction.

Architecture utilizes a bifurcated design: The LSTM layer handles stochastic thermal variance, while the Agentic Logic applies deterministic sustainability constraints to optimize routing in real-time.

Validation & Sustainability Impact

66%

Spoilage Reduction

By transitioning from reactive monitoring to **Agentic AI Intervention**, the framework achieves high-fidelity waste mitigation:

- ✓ **Zero-Touch Decisions:** Autonomous rerouting without driver intervention.
- ✓ **Carbon Savings:** Direct correlation between spoilage reduction and CO2 mitigation.
- ✓ **National Resilience:** Strengthening pharma and food security.